

Lab Tasks — Entity-Relationship Diagrams

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DATE

19 Mar 2026



Part 1 — ERD Drawing

1 task · Draw the Library System ERD on paper and upload a photo

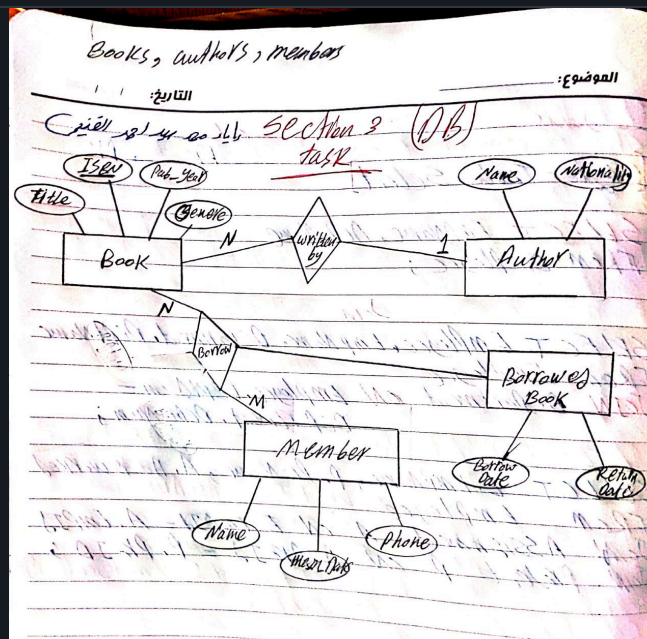
T 1.1

Read the **Library System** scenario below, then draw the complete **Chen notation ERD** on paper. Include all entities, attributes (with key attributes underlined), relationships, and cardinality labels.

A public library manages **books**, **authors**, and **members**. Each book has a title, ISBN, publication year, and genre. Each author has a name and nationality. A book is written by one author, but an author can write many books. Library members have a name, membership date, and phone. Members can borrow multiple books, and a book can be borrowed by different members over time. Each borrowing records the borrow date and return date.

- 💡 Use the 7-step process from the guide: (1) Read → (2) Highlight nouns → (3) Highlight verbs → (4) List attributes → (5) Identify keys → (6) Determine cardinality → (7) Draw & validate. You should find 3 strong entities + 1 junction entity.

PHOTO — YOUR HAND-DRAWN ERD



Part 2 — Entity & Attribute Extraction

4 tasks · Identify entities and their attributes from the Library scenario

T 2.1

Fill in the **entities** and their **attributes** for the Library System. Mark which attribute is the **Primary Key (PK)**.

ENTITY 1 — NAME:

ATTRIBUTE	TYPE	PK?
AuthName	NVARCHAR(100)	NO
AuthNationality	NVARCHAR(150)	No
...		

ENTITY 2 — NAME:

ATTRIBUTE	TYPE	PK?
ISBN	INT	Yes
BookTitle	NVARCHAR(200)	NO
PubYear	DATE	NO
BookGen	NVARCHAR(100)	NO

ENTITY 3 — NAME:

ATTRIBUTE	TYPE	PK?
MemName	NVARCHAR(100)	NO
MemDate	DATETIME	NO
MemPhone	NVARCHAR(20)	NO

ENTITY 4 (JUNCTION) — NAME: **BorrowingBooks**

ATTRIBUTE	TYPE	PK?
BorrowDate	DATETIME	NO
ReturnDate	DATETIME	NO

T 2.2 One of the entities you identified above acts as a **junction (bridge) entity**. Which entity is it, and why does it need to be a **separate entity** rather than just a relationship?

The junction entity is (Borrowing Book). It is a separate entity because it resolves the Many-to-Many relationship between Members and Books and allows the system to store specific transaction data like Borrow Date and Return Date



Part 3 — Relationships & Cardinality

3 tasks · Identify relationships and determine cardinality

T 3.1 Match each relationship to its **cardinality**:

Author → Book ("*writes*")

1:N



Member ↔ Book ("*borrow*s")

M:N



T 3.2 For the **1:N** relationship (Author → Book), which table gets the **Foreign Key**?

- ☐ The Authors table
- ☒ The Books table
- ☐ Both tables
- ☐ A new junction table

T 3.3 How is the M:N relationship between Member and Book **resolved**? What is the junction entity called, and what attributes does it have?

The M:N relationship is resolved by creating a junction entity called Borrowing. This entity acts as a bridge that links Members and Books, storing specific transaction attributes like Borrow Date and Return Date.



Part 4 — ERD → Table Mapping (SQL)

2 tasks · Write CREATE TABLE statements for the Library System

T 4.1

Write the **CREATE TABLE** SQL statements for all 4 tables in the Library System: **Authors**, **Books**, **Members**, and **Borrowings**. Include data types, PRIMARY KEY, and FOREIGN KEY constraints.



Remember: 1:N → FK in many side. M:N → junction table with composite PK. Use INT for IDs, VARCHAR for text, DATE for dates.

YOUR SQL

```
CREATE TABLE Author(
```

```
    AuthID          INT          PRIMARY KEY,  
    AuthName        NVARCHAR(100) NOT NULL,  
    AuthNationality NVARCHAR(150)
```

```
);
```

```
CREATE TABLE Book(
```

```
    ISBN          INT          PRIMARY KEY,  
    BookTitle      NVARCHAR(200) NOT NULL,  
    PubYear        DATE,  
    BookGen        NVARCHAR(100),  
    AuthID         INT          FOREIGN KEY REFERENCES  
    Author(AuthID)
```

```
);
```

```
CREATE TABLE Members(
```

```
    MemID          INT          PRIMARY KEY,  
    MemName        NVARCHAR(100) NOT NULL,  
    MemDate        DATETIME     DEFAULT GETDATE(),  
    MemPhone       NVARCHAR(20)
```

```
);
```

```
CREATE TABLE BorrowingBooks(
```

```
    ISBN          INT          FOREIGN KEY REFERENCES  
    Book(ISBN),  
    MemID         INT          FOREIGN KEY REFERENCES  
    Members(MemID),  
    BorrowDate    DATETIME     NOT NULL,  
    ReturnDate    DATETIME     NOT NULL,  
    PRIMARY KEY(ISBN, MemID, BorrowDate)
```

```
);
```

T 4.2

Explain in your own words: **Why does the Borrowings table have a composite primary key?** What columns make up that composite PK?

The Borrowing table uses a composite primary key to uniquely identify each transaction, as a single Book or Member ID is not enough to distinguish between different rental events. The composite PK is made up of ISBN, MemID, and BorrowDate to ensure the same member can borrow the same book at different times.